

TS Question Paper 2017**Class 12 Chemistry,****Paper-II****Time : 3 hours****Max. Marks : 60****Note: Read the following instructions carefully:**

- 1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.**
- 2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.**
- 3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each questions carries four marks. Every answer may be limited to 75 words.**
- 4) In Section-‘C’, Questions from Sr.Nos.19 to 21 are of” Long answer type”. Each questions carries eight marks. Every answer may be limited to 300 words.**
- 5) Draw labeled diagram, wherever necessary for questions in section-‘B’ and ‘C’.**

SECTION-A**(10 × 2 = 20)**

Note: Answer all the questions.

1. What is metallic corrosion? Give an example.

Ans:- Metallic corrosion occurs when metal atoms are oxidized and subsequently leave the metal lattice as ions. The oxidation reaction of the metal is referred to as the anodic half reaction. The areas on a metal surface where oxidation occurs are referred to as anodes.

Example- Steel rivets in alloy 400 sheet exposed to seawater.

2. What is reverse osmosis? What is practical utility?

Ans-The direction of osmosis can be reversed if a pressure larger than the osmotic pressure is applied to the solution side. The pure solvent flows out of the solution through the semipermeable membrane. This phenomenon is called reverse osmosis.

In other words, Reverse Osmosis is a water purification technology that uses a semipermeable membrane to remove ions, molecules and larger particles from drinking water.

The practical utility of reverse osmosis is the desalination of seawater. Reverse osmosis is used in drinking water purification from seawater, removing the salt and other effluent materials from the water molecules.

3. Give the composition of the following alloys:

- a) Brass b) German silver

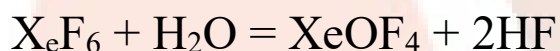
Ans- a) Copper-67%, Zinc-33%

b) Copper-50% to 61%, Zinc- 19% to 17%, nickel - 30% to 21%

4. How is XeOF₄ prepared? Describe its molecular shape.

Ans:-Preparation of XeOF₄.

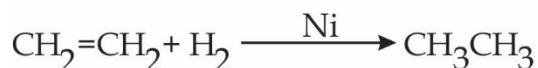
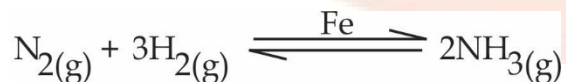
It can be prepared using XeF₆. The partial hydrolysis of XeF₆ gives XeOF₄.



XeOF₄ has octahedral geometry due to sp₃d₂ hybridisation. One trans position is occupied by a lone pair giving pyramidal shape to the molecule. Hence the molecular shape of XeOF₄ is square pyramidal.

5. Give two reactions in which transition metals or their compounds act as catalysts.

Ans:-



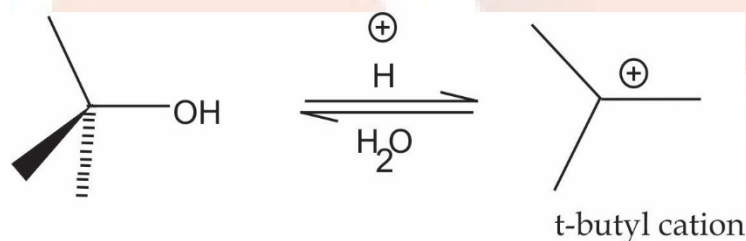
6. What are ambident nucleophiles?

Ans- Ambient nucleophiles are those nucleophiles which contain 2 nucleophilic centres. In other words, two sites of negative charge and excess electrons. A good example of an ambient is the cyanide ion CN^- here the nucleophilic centres are C- and :N:

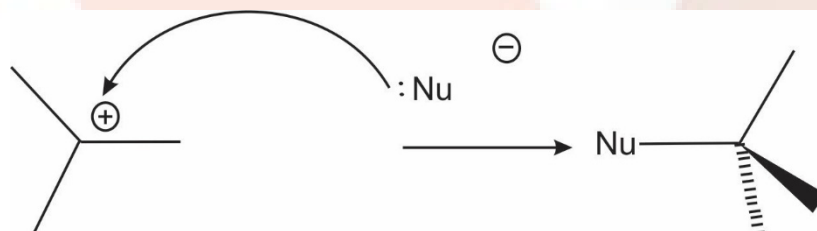
7. What is stereochemical result of $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$ reactions?

Ans- $\text{S}_\text{N}1$:- Unimolecular Nucleophilic Substitution.

Step # 1 : Formation of the cation (rate determining step)

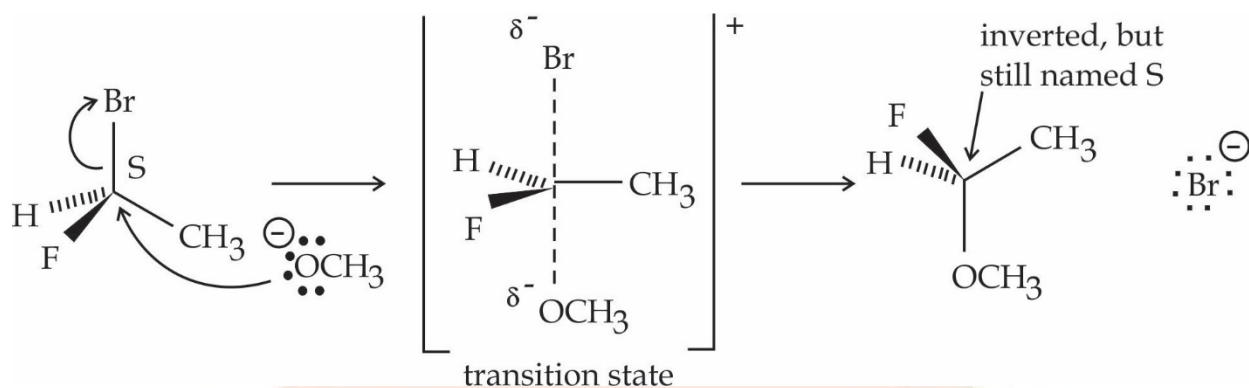


Step # 2 : Reaction of the carbocation and the nucleophile (fast step)

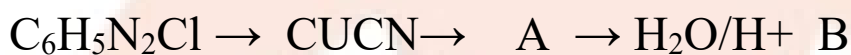


$\text{S}_\text{N}2$:- Bimolecular Nucl. Sub.:-

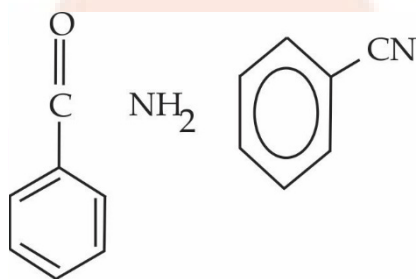
They differ primarily in the mechanism, which is why they are favored in different types of solvents.



8. Give the structures of A and B.



A- Benzonitrile

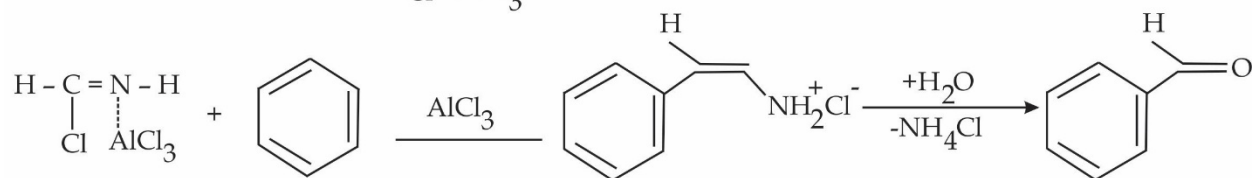
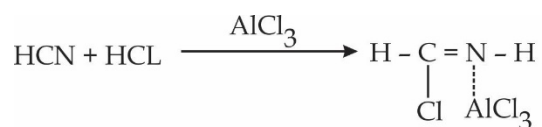


B- Benzamide

9. Explain Gatterman Reaction.

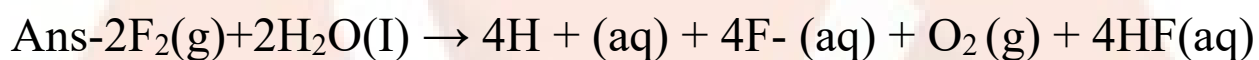
Ans-In this reaction, the benzene is added with Hydrocyanic acid and Friedel craft's catalyst AlCl_3 to form benzene diazonium chloride which on further hydrolysis produces as aldehyde. In such reaction, copper acts as a catalyst.

The reaction is :



This type of reaction is also used in synthesis of halo arenes also where Cu-powder is used as catalyst.

10. Write the reactions of F₂ with water.



SECTION- B**(6 × 4 = 24)****Note: Answer Any Six questions.****11. Define molarity Calculate the molarity of a solution containing 5g of NaOH in 500ml of solution.**

Ans- Molarity describes the number of moles of the substance per liter of liquid. In chemistry, a mole is a unit of measurement for a chemical substance. The scientific noun molarity describes the concentration of a substance as measured by the number of moles divided by the volume of the solution it's in.

Molar mass of NaOH=40g per mole

no. of moles in 5g of NaOH= $5/40=0.125$

volume of solution=450mL=0.45L

molarity of solution=no.of moles of solute/volume of solution

$$= 0.125/0.45$$

$$= 0.277777777778=0.28(\text{approx.})$$

12. Derive Bragg's Equation.

Ans- Let's consider a single monochromatic light incident on a point with difference d, at angle theta.

Point A and C are on same plane and point B is on another plane. Together point ABCC' formed a quadrilateral.

The path difference is, $(AB-BC) - (AC')$ this path difference is

equal to the any integer value of wavelength.,

$$(AB+BC)-(AC')= n\lambda$$

Therefore, $AB=BC=d/\sin \theta$ or,

$$AC=2d/\tan \theta$$

$$AC'=AC.\cos \theta = 2d/\tan \theta.\cos \theta = (2d/\sin \theta.\cos \theta)\cos \theta = 2d/\sin \theta.\cos^2 \theta$$

If we put these values together , we get

$$n\lambda = 2d/\sin \theta - 2d/\sin \theta.\cos^2 \theta = 2d/\sin \theta (1 - \cos^2 \theta) = 2d/\sin \theta.\sin^2 \theta,$$

after simplifies we get,

$$n\lambda = 2d \sin \theta$$

13. Give any four differences between physisorption and chemisorptions.

Ans:

Physisorption	Chemisorptions
1.It arises due to Vanderwaal Forces	1.It arises due to Chemical bond formation.
2.It is reversible.	2.It is irreversible.
3.It is not specific in nature.	3.It is highly specific in nature.

4. Enthalpy of Adsorption is low (20-40 kJ/mol)	4. Enthalpy of adsorption is high (80-240 kJ/mol).
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14. Explain briefly the extraction of aluminium from bauxite.

Ans:- Extraction of aluminium

Aluminium is the most abundant metal on Earth. Despite this, it is expensive, largely because of the amount of electricity used up in the extraction process.

Aluminium ore is called bauxite. The bauxite is purified to yield a white powder, aluminium oxide, from which aluminium can be extracted.

Aluminium metal forms at the negative electrode and sinks to the bottom of the tank, where it is tapped off.

Oxygen forms at the positive electrodes. This oxygen reacts with the carbon of the positive electrodes, forming carbon dioxide, and they gradually burn away. Consequently, the positive electrodes have to be replaced frequently, which adds to the cost of the process.

15. Write the formulas for the following coordination compounds.

a) Tetraammineaquachloro cobalt(III)chloride

b) Potassium tetrahydroxozincate(II)

c) Potassium trioxalatoaluminate(III)

d) Tetracarbonylnickel(0)

Ans-a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$

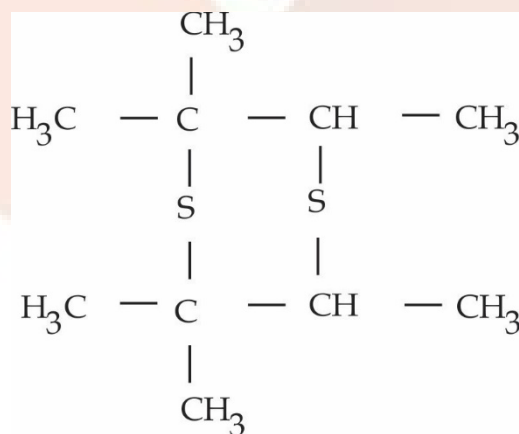
b) $\text{K}_2[\text{Zn}(\text{OH})_4]$

c) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$

d) $\text{Ni}(\text{CO})_4$

16. Explain the purpose of vulcanization of rubber.

Ans- Natural rubber becomes soft at high temperature and brittle at low temperatures. To improve physical properties of rubber, vulcanization is carried out. This process consists of heating a mixture of raw rubber with sulphur and an appropriate additive at a temperature range between 373K and 415K. On vulcanization, sulphur forms cross-links at the reactive sites of double bonds and thus the rubber gets stiffened.



Sulphur cross-link in vulcanized rubber

17. Give the sources of the following vitamins and the name the diseases caused by their deficiency.

- a) A b) D c) E d) K

Ans:- Vitamins Sources Diseases caused by deficiency

- a) A Apricots liver disorders, hematopoiesis
b) D Cheese, Liver soft bones, skeletal deformities
c) E Nuts, Seeds Disorientation and vision problem
d) K Fish, Eggs Bleeding, hemorrhagic disease of the newborn

18. Write notes on antiseptics and disinfectants.

Ans- Antiseptics:

These drugs are applied to living tissues to kill the bacteria and to stop their growth in wound, thus preventing its infection. Antiseptics do not heal the wound but prevent the wound from getting infected. Dettol (a mixture of terpineol and chloroxylenol) is an antiseptic. Other antiseptics are tincture iodine, bithional, iodoform, boric acid, phenol (dilute solution) etc.

Disinfectants:

Disinfectants are applied to non-living objects to kill the microorganisms, used in public health sanitation, floors, to streamline instruments, etc. Examples include chlorine, sulphur dioxide, phenol (concentrated solution).

SECTION – C

 $(2 \times 18 = 16)$

Note : Answer ANY TWO questions.

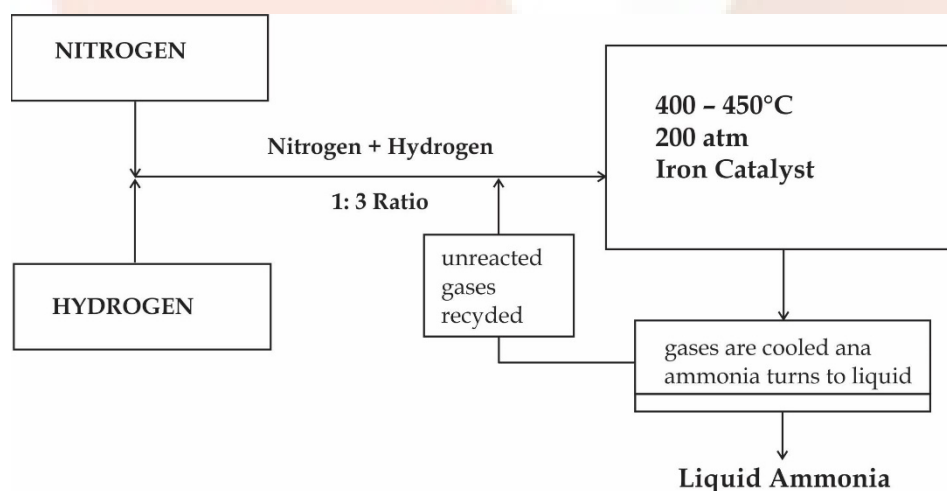
19.a) How is Ammonia manufactured by haber's Process with a neat diagram?

Ans:- In the Haber process, “the atmospheric nitrogen (N_2) is converted to ammonia (NH_3) by reacting it with hydrogen (H_2)”. Here a metal catalyst is used and high temperatures and pressures are maintained.

The raw materials for the process are

- Air, which supplies the nitrogen.
- Natural gas and water which supply the hydrogen and the energy needed to heat the reactants.

Let us take a look at the diagram below.



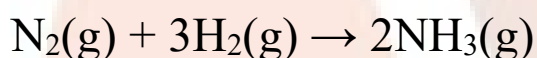
1. As per the diagram, in the Haber process, we take nitrogen gas from the air and combine it with hydrogen atom obtained

from natural gas in the ratio 1:3 by volume.

2. The gases are passed through four beds of catalyst, with cooling takes place in each pass. This is done to maintain equilibrium constant.

3. The process also involves steps like shift conversion, carbon dioxide removal, steam reforming, and methanation.

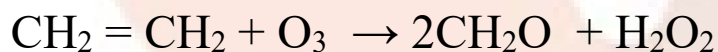
4. In the final stage of the process, the ammonia gas is cooled down to form a liquid solution which is then collected and stored in storage containers.



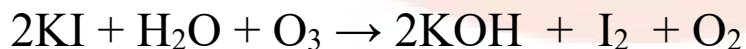
b) Explain the reaction of the following with ozone.

a) C_2H_4 b) KI c) Hg d) PbS

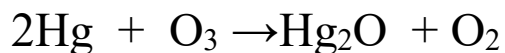
Ans:-i) C_2H_4 :-Ethylene reacts with ozone to produce ozonide which is decomposed by Zn dust and water to form formaldehyde.



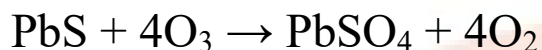
ii) KI :- Moist potassium Iodide with ozone to produce potassium hydroxide, iodine and oxygen.



iii) Hg :- Mercury reacts with ozone to form an oxide. Meniscus of mercury is lost and mercurous oxide sticks to the glass. It is known as tailing of mercury.



iv) PbS:- Lead Sulfide react with ozone to produce lead sulphate and oxygen at room temperature.



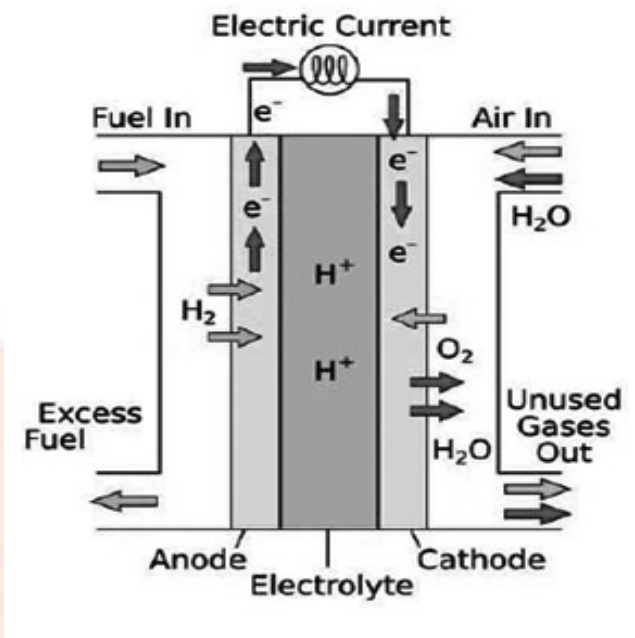
20.a) What are the fuel cells? How are they different from Galvanic cells? Give the construction of H_2 and O_2 fuel cells.

Ans:-A fuel cell is a device that generates electrical power through a chemical reaction by converting a fuel (hydrogen) into electricity. Although fuel cells and batteries are both considered electrochemical cells and consist of similar structure, fuel cells require a continuous source of fuel and oxygen to run; similar to how an internal combustion engine needs a continuous flow of gasoline or diesel.

A battery is a galvanic cell that contains all the reactants needed to produce electricity.

A fuel cell is a galvanic cell that requires a constant external supply of one or more reactants to generate electricity.

Fuel Cell



AT THE ANODE (-electrode)

OXIDATION TAKES PLACE

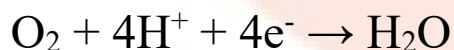


These H⁺ ions move through the electrolyte

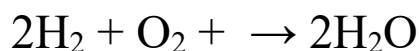
(often a polymer or membrane)

AT THE CATHODE (+ electrode)

REDUCTION TAKE PLACE



OVERALL REACTION



b) What is molecularity of a reactions?How it is different from the order of a reaction?Name one bimolecular and one

trimolecular gaseous reactions.

Ans: Molecularity. ... The molecularity of a reaction is defined as the number of molecules or ions that participate in the rate determining step. A mechanism in which two reacting species combine in the transition state of the rate-determining step is called bimolecular.

1. The order of a reaction reduces if any reactant is in excess however it is not so in the case of molecularity .
2. Further the order is decided on the basis of rate determining step it is not so in case of molecularity .
3. Molecularity is not defined for a complex reaction or multi-step reaction however order is.
4. Molecularity always has non negative integral values but order can take any real values

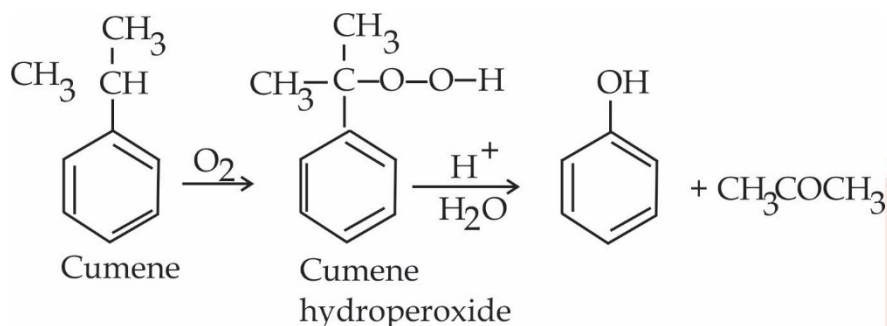
Bimolecular reaction- $2\text{NOCl} \rightarrow 2\text{NO}_2(\text{g}) + \text{CO}_2(\text{g})$

Trimolecular:- $2\text{NO}(\text{g}) + \text{O}_2 \rightarrow 2\text{NO}_2(\text{g})$

21.a) Explain the preparation of phenol from cumene.

Ans:-Cumene is an organic compound obtained by Friedel-Crafts alkylation of benzene with propylene. Upon oxidation of cumene (isopropylbenzene) in the presence of air, cumene hydroperoxide is obtained. Upon further treatment of cumene hydroperoxide with dilute acid, phenols are obtained. Acetone is also produced as one of the by-products of this reaction in large

quantities. Hence, phenols prepared by these methods need purifications.

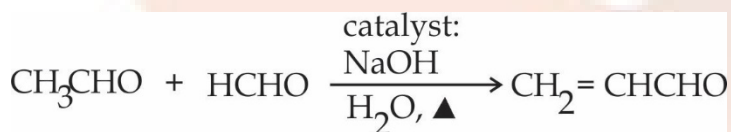


b) Describe the following:

- i) Cross aldol condensation
- ii) Decarboxylation

Ans:-i) A crossed aldol condensation uses two different aldehyde and/or ketone reactants. Such reactions usually give a mixture of multiple condensation products, because there are two or more possible enolate nucleophiles, and two different carbonyl electrophiles.

For example, ethanol and methanol react to give four products.



ii) Decarboxylation:- Decarboxylation is a chemical reaction that removes a carboxyl group and releases carbon dioxide. Usually, decarboxylation refers to a reaction of carboxylic acids, removing a carbon atom from a carbon chain

